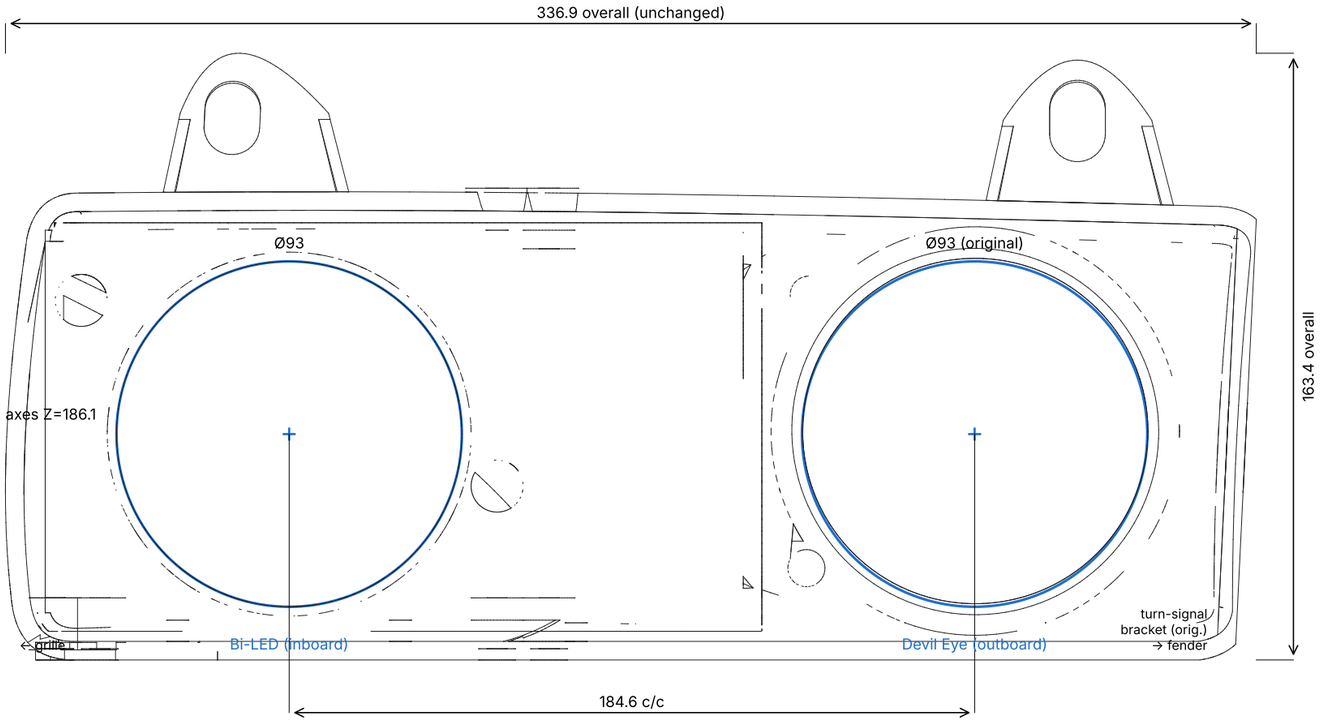
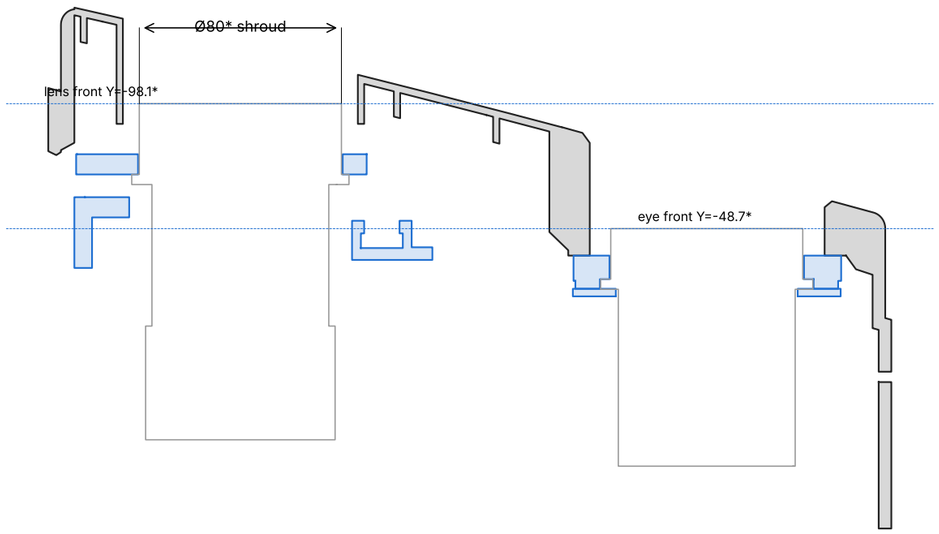
Sheet 1 — Assembly overview & key dimensions

BMW E36 modular headlight — LEFT (Vänster) shown, RIGHT = mirror. Units mm. Frame: original housing STL (-Y forward, +Z up, +X outboard). Black = housing (modified original), blue = new printed parts, grey = ASSUMED module envelopes, red = purchased hardware.

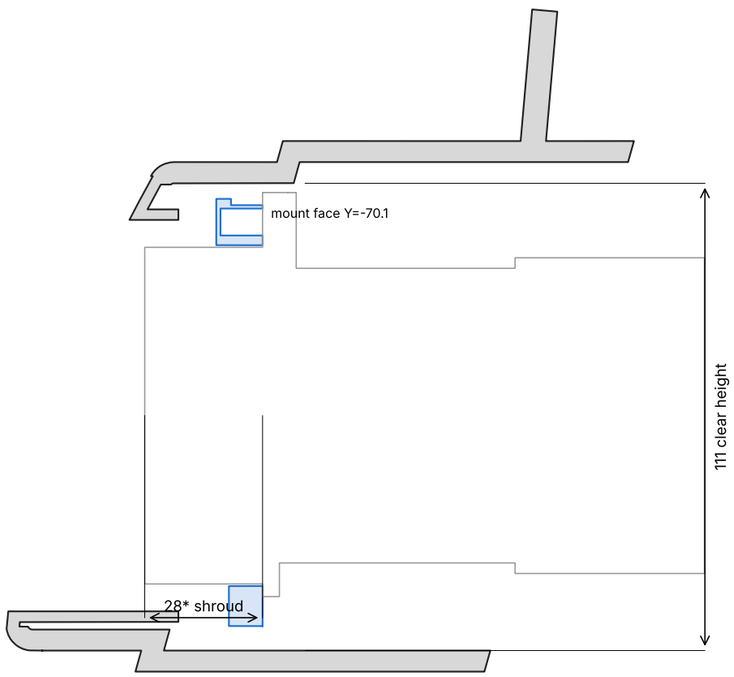
Front view (from -Y) — outline + section Y=-60



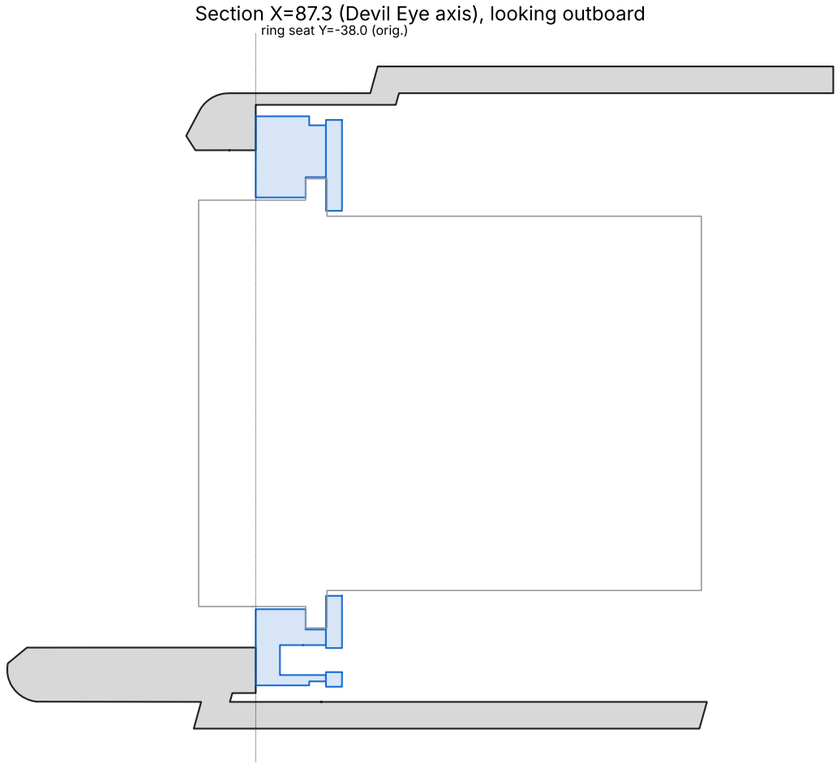
Plan section Z=186.1 (looking down)



Section X=-97.3 (projector axis), looking outboard



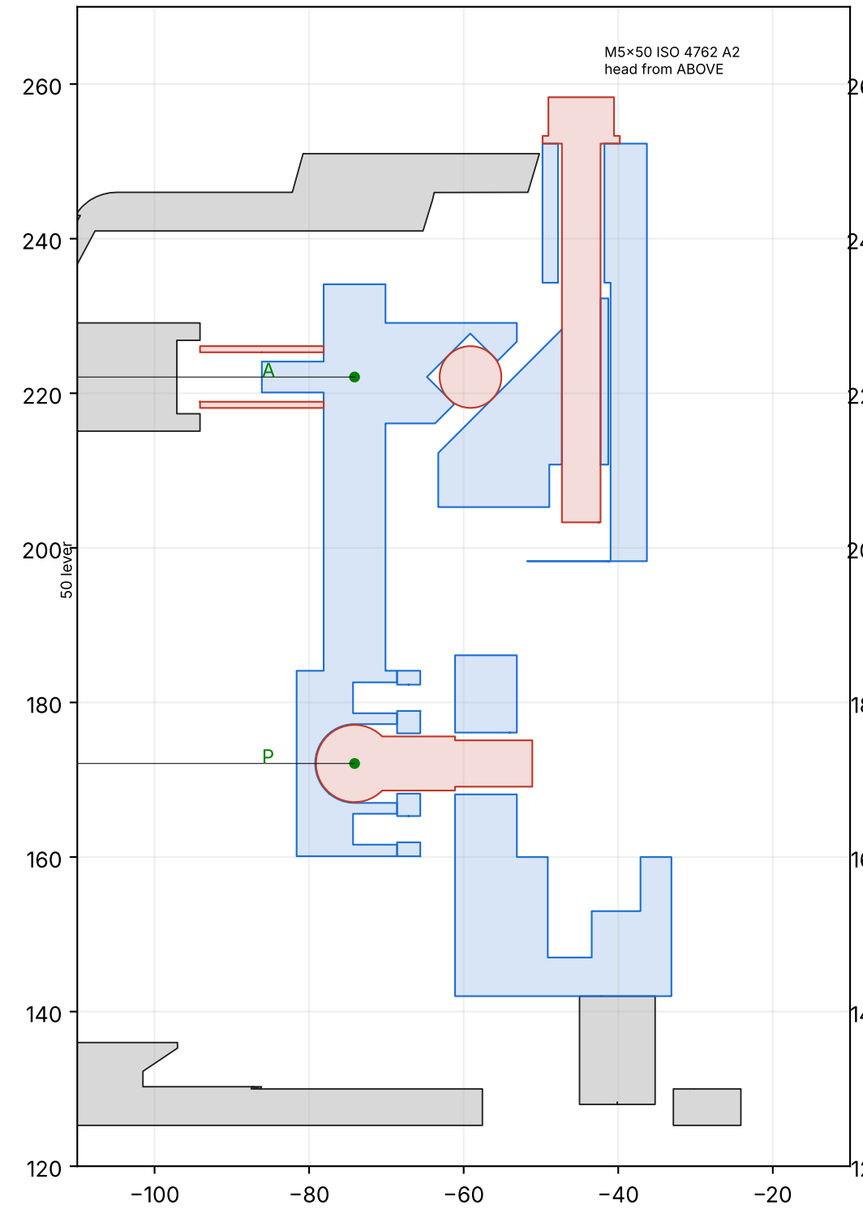
Section X=87.3 (Devil Eye axis), looking outboard



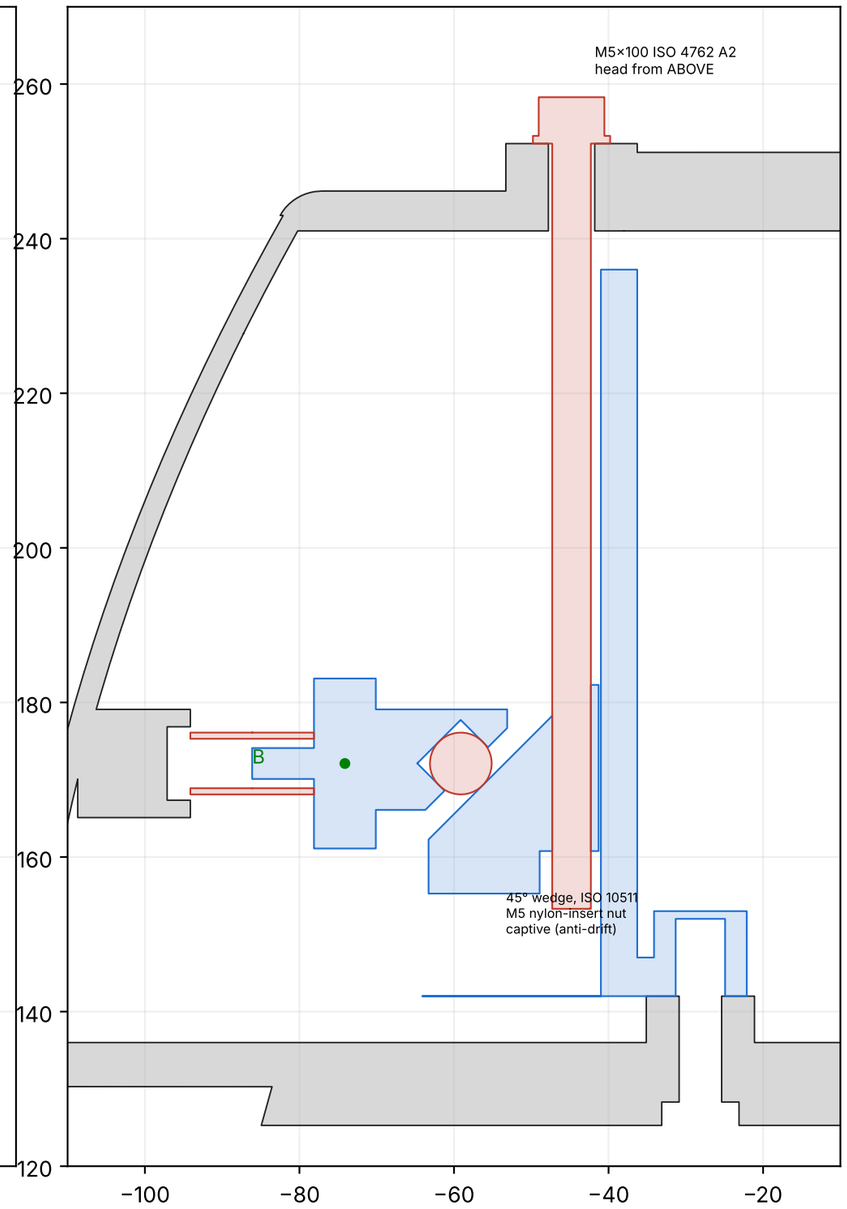
Sheet 2 — Aiming mechanism (3-point, top-access wedge adjusters)

BMW E36 modular headlight — LEFT (Vänster) shown, RIGHT = mirror. Units mm. Frame: original housing STL (-Y forward, +Z up, +X outboard). Black = housing (modified original), blue = new printed parts, grey = ASSUMED module envelopes, red = purchased hardware.

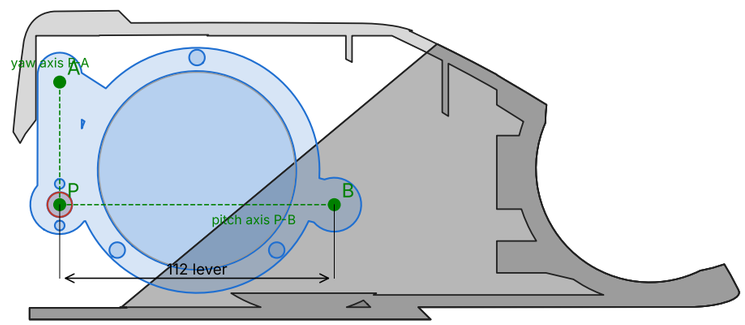
Section X=-153.3 through P and A (pitch)



Section X=-41.3 through B (yaw)



Front view of carrier (section Y=-74.1)



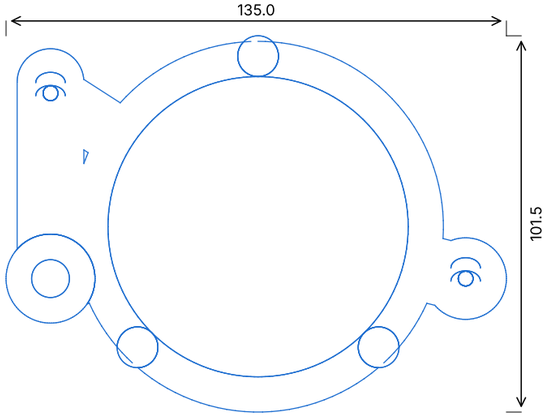

```
Kinematics / validation (validate.py, real booleans):
  pitch: 0.917°/turn, yaw: 0.409°/turn
  range checked ±3° both axes, 3x3 sweep: CLEAR
  min shroud-to-aperture clearance over sweep: 4.74 mm*
  wedge travel needed A/B: 2.62 / 5.87 mm (ramp ±7)
  rear removal: projector clear, devil eye clear

Preload: 2 compression springs Ø8 x 25 free / 16 installed
  between skin posts and carrier spigots (in line with A, B)
  push the carrier rearward onto the wedges; the ball load
  hangs each screw in tension from its head (no drift path).
Pivot P: DIN 71803 Ø10 ball stud (M6) in a printed socket,
  closed by the 2xM3 pivot cap. Followers: Ø8 steel balls.
* depends on ASSUMED module dimensions — re-run after measuring.
```

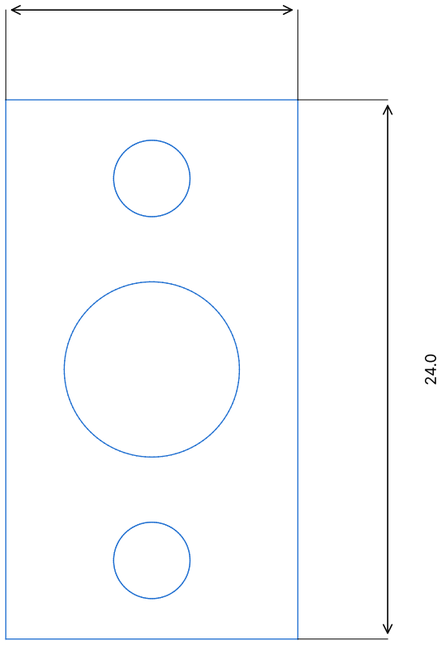
Sheet 3 — New printed parts (installed orientation, bounding dims)

BMW E36 modular headlight — LEFT (Vänster) shown, RIGHT = mirror. Units mm. Frame: original housing STL (-Y forward, +Z up, +X outboard). Black = housing (modified original), blue = new printed parts, grey = ASSUMED module envelopes, red = purchased hardware.

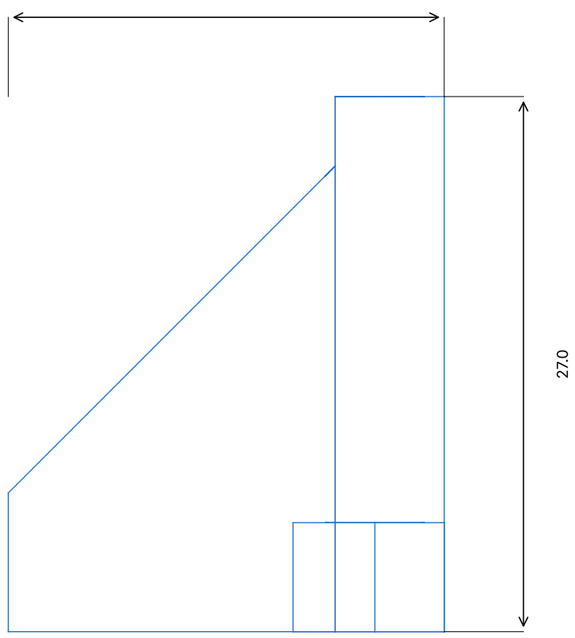
projector_carrier (135.0 x 33.0 x 101.5, 39.8 cm³)
view along Y



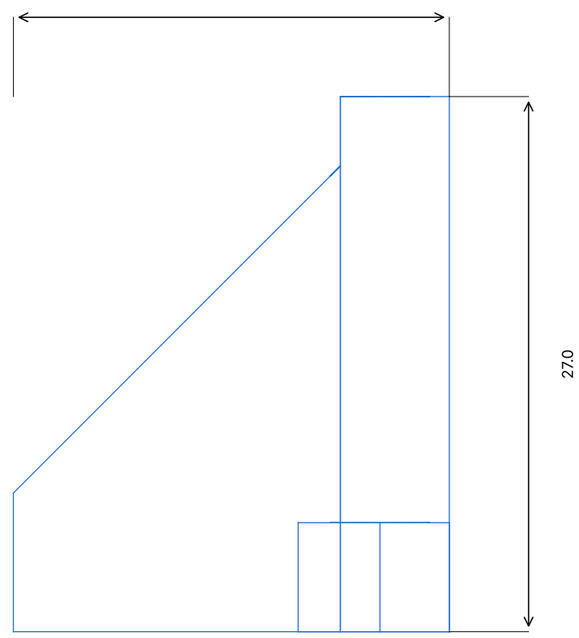
pivot_cap (13.0 x 3.0 x 24.0, 0.7 cm³)
view along Y



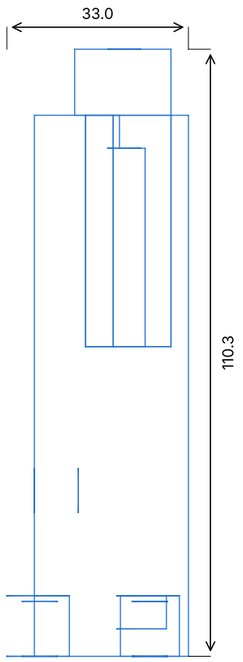
wedge_A (16.0 x 22.0 x 27.0, 4.7 cm³)
view along X



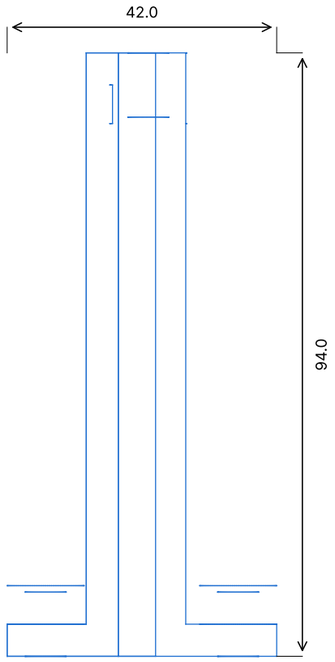
wedge_B (16.0 x 22.0 x 27.0, 4.7 cm³)
view along X



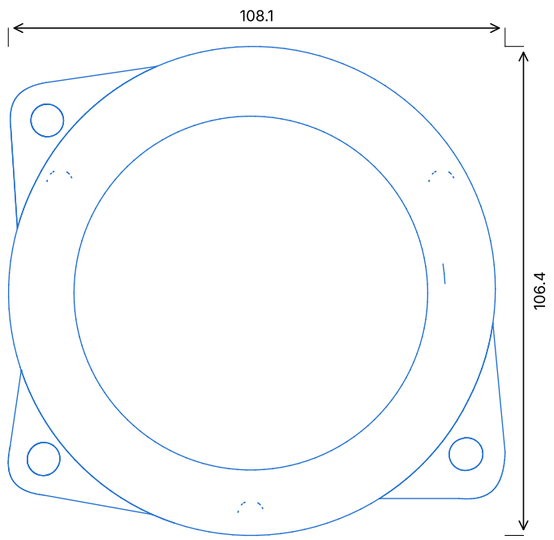
inboard_bracket (25.0 x 33.0 x 110.3, 33.0 cm³)
view along X



outboard_pillar (31.8 × 42.0 x 94.0, 27.5 cm³)
view along X



devil_adapter (108.1 × 13.1 × 106.4, 54.8 cm³)
view along Y



devil_clamp (106.0 x 3.0 x 106.0, 14.1 cm³)
view along Y

